

COMP 3311: Database Management Systems

Lab 6 Exercise: SQL DML Modification Statements and Basic SQL DDL Statements

WHAT TO DO

1. **Download** the zipped folder **Lab6ExerciseFiles.zip** from the *Lab 6* entry of the *Lab Assignments* section on Canvas to the desktop and **unzip** it. The folder contains three script files **lab6db.sql**, **lab6AddConstraints.sql** and **lab6TestConstraints.sql**.
 2. **Place** your **insertMyself.sql** script file **inside** the **Lab6ExerciseFiles** folder.
 3. **Execute** the **lab6db.sql** script file in **Oracle SQL Developer**.
 4. **Edit** the **lab6AddConstraints.sql** script file by editing the **alter table** statements to do the following.
 - a. For the **Student**, **Course** and **EnrollsIn** tables, add foreign key referential integrity constraints.
 - b. For the **AcademicUnit** table, add a constraint to check that **unitId** has only a value in the set {BUS, COMP, ELEC, HUMA, MATH}.
 - c. For the **Student** table, add constraints to
 - i. check that **studentId** and **phoneNo** have exactly 8 digits only;
 - ii. check that **cga** has a value between 0 and 4;
 - iii. check that **admitYear** begins with a 2 and has exactly 4 digits only; and
 - iv. enforce the uniqueness of **email** values.
 - d. For the **Course** table, add a constraint to check that **courseld** follows the pattern of exactly four uppercase letters followed by exactly four digits.
 - e. For the **EnrollsIn** table, add a constraint to check that **grade** has a value between 0 and 100.
- Note:** As you construct the constraints, you can test each one **individually** for correctness by first executing it in your **lab6AddConstraints.sql** script file and then executing the corresponding test query in the **lab6TestConstraints.sql** script file. If a constraint is not correctly defined, then executing its corresponding test query will insert one or more records into a table that violates the constraint. To remove the incorrectly inserted records you will need to *either* delete them using appropriate delete statements *or* execute the **lab6db.sql** script file again **before** executing the constraint's test query.
5. **Test** your constraints to ensure that they execute without errors and obtain the expected result as shown below.
 6. When all your constraints are correctly specified, **execute** the **lab6TestConstraints.sql** script file in **Oracle SQL Developer** to create the required **lab6TestConstraintsResults.txt** submission file.

WHAT TO SUBMIT

1. Your modified **lab6AddConstraints.sql** script file containing the completed **alter table** statements.
2. A text file named **lab6TestConstraintsResults.txt** containing the result of executing the **lab6TestConstraints.sql** script file. The **lab6TestConstraintsResults.txt** file can be created by saving the **SQL Developer** Script Output tab to a file with this name. The expected result of executing the **lab6TestConstraints.sql** script file is shown below.

This is an individual exercise. Each student must submit their own exercise files.

HOW TO SUBMIT

By 6:00 p.m. tomorrow, upload your **lab6AddConstraints.sql** script file and **lab6TestConstraintsResults.txt** text file to Canvas.

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EXPECTED RESULT OF EXECUTING THE LAB6TESTCONSTRAINTS.SQL SCRIPT FILE

<<<<----- STUDENTS WHOSE AVERAGE GRADE IS >90 ----->>>>

STUDENTI	FIRSTNAME	LASTNAME	TRUNC(AVG(GRADE),2)
15000655	Steve	Jobs	91.47
11111111	Typical	Student	91.18
28918856	Alan	Turing	91.1

Your student record should be shown in the result of this query.

<<<<----- TESTS FOR REFERENTIAL INTEGRITY CONSTRAINTS ----->>>>

*** 1. REFERENTIAL INTEGRITY on Student table: No matching unitId in AcademicUnit table.

Error starting at line : 26 in command -

insert into Student values ('22222222', '*** Student ***', 'RI: No matching dept id', 'afong', '22223334', '2023', 0.00, 'PHYS')

Error report -

ORA-02291: integrity constraint (COMP3311STU000.SYS_C00892216) violated - parent key not found

<https://docs.oracle.com/error-help/db/ora-02291/>

More Details :

<https://docs.oracle.com/error-help/db/ora-02291/>

*** 2. REFERENTIAL INTEGRITY on Student table: Should delete 1 row and select 0 rows.

1 row deleted.

0 rows selected.

*** 3. REFERENTIAL INTEGRITY on Course table: No matching unit id in AcademicUnit table.

Error starting at line : 36 in command -

insert into Course values ('PHYS4311', '*** Course ***', 3, 'RI: No matching dept id', 'PHYS')

Error report -

ORA-02290: check constraint (COMP3311STU000.CHK_COURSEID) violated

<https://docs.oracle.com/error-help/db/ora-02290/>

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*** 4. REFERENTIAL INTEGRITY on EnrollsIn table: No matching course id in Course table.

Error starting at line : 41 in command -

insert into EnrollsIn values ('13456789', 'PHYS4311', 80.6)

Error report -

ORA-02291: integrity constraint (COMP3311STU000.SYS_C00892219) violated - parent key not found

<https://docs.oracle.com/error-help/db/ora-02291/>

More Details :

<https://docs.oracle.com/error-help/db/ora-02291/>

*** 5. REFERENTIAL INTEGRITY on EnrollsIn table: No matching student id in Student table.

Error starting at line : 45 in command -

insert into EnrollsIn values ('12345678', 'COMP3311', 75.6)

Error report -

ORA-02291: integrity constraint (COMP3311STU000.SYS_C00892219) violated - parent key not found

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<<<<----- TESTS FOR CHECK CONSTRAINTS ----->>>>

*** 6. CHECK CONSTRAINT on AcademicUnit table: unitId must be one of COMP, ELEC, FINA, HUMA, MATH or MGMT.

Error starting at line : 52 in command -

insert into AcademicUnit values ('DEPT','*** AcademicUnit table: CHK: deptId','0000')

Error report -

ORA-02290: check constraint (COMP3311STU000.CHK_UNITID) violated

<https://docs.oracle.com/error-help/db/ora-02290/>

More Details :

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*** 7. CHECK CONSTRAINT on Student table: studentId must be all digits.

Error starting at line : 57 in command -

insert into Student values ('234567*9', '*** Student table:', 'CHK: studentId', 'afong', '2223334', '2023', 0.00, 'COMP')

Error report -

ORA-02290: check constraint (COMP3311STU000.CHK_STUDENTID) violated

<https://docs.oracle.com/error-help/db/ora-02290/>

More Details :

<https://docs.oracle.com/error-help/db/ora-02290/>

*** 8. CHECK CONSTRAINT on Student table: studentId must be exactly 8 digits.

Error starting at line : 62 in command -

insert into Student values ('234567', '*** Student table:', 'CHK: studentId', 'bfong', '2223334', '2023', 0.00, 'COMP')

Error report -

ORA-02290: check constraint (COMP3311STU000.CHK_STUDENTID) violated

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More Details :

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*** 9. CHECK CONSTRAINT on Student table: phoneNo must be all digits.

Error starting at line : 67 in command -

insert into Student values ('33333333', '*** Student table:', 'CHK: studentId', 'cfong', '222a3334', '2023', 0.00, 'COMP')

Error report -

ORA-02290: check constraint (COMP3311STU000.CHK_PHONENO) violated

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*** 10. CHECK CONSTRAINT on Student table: phoneNo must be exactly 8 digits.

Error starting at line : 72 in command -

insert into Student values ('44444444', '*** Student table:', 'CHK: studentId', 'dfong', '2222', '2023', 0.00, 'COMP')

Error report -

ORA-02290: check constraint (COMP3311STU000.CHK_PHONENO) violated

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More Details :

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*** 11. CHECK CONSTRAINT on Student table: cga must be between 0 and 4.

Error starting at line : 77 in command -

insert into Student values ('55555555', '*** Student table:', 'CHK: cga', 'ali', '25524334', '2023', 4.50, 'COMP')

Error report -

ORA-02290: check constraint (COMP3311STU000.CHK_CGA) violated

<https://docs.oracle.com/error-help/db/ora-02290/>

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*** 12. CHECK CONSTRAINT on Student table: admitYear must be all digits.

Error starting at line : 82 in command -

insert into Student values ('77777777', '*** Student table:', 'CHK: admitYear', 'bli', '25524334', '200a', 4.50, 'COMP')

Error report -

ORA-02290: check constraint (COMP3311STU000.CHK_ADMITYEAR) violated

<https://docs.oracle.com/error-help/db/ora-02290/>

More Details :

<https://docs.oracle.com/error-help/db/ora-02290/>

*** 13. CHECK CONSTRAINT on Student table: admitYear has exactly 4 digits.

Error starting at line : 87 in command -

insert into Student values ('88888888', '*** Student table:', 'CHK: admitYear', 'cli', '25524334', '200', 4.50, 'COMP')

Error report -

ORA-02290: check constraint (COMP3311STU000.CHK_ADMITYEAR) violated

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More Details :

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*** 14. CHECK CONSTRAINT on Student table: admitYear begins with a 2.

Error starting at line : 92 in command -

insert into Student values ('99999999', '*** Student table:', 'CHK: admitYear', 'dli', '25524334', '1223', 4.50, 'COMP')

Error report -

ORA-02290: check constraint (COMP3311STU000.CHK_ADMITYEAR) violated

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More Details :

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*** 15. CHECK CONSTRAINT on Course table: course id must be exactly four uppercase letters, a space and four digits.

Error starting at line : 97 in command -

insert into Course values ('comp 3311', '** Course Table: CHK: courseId', 3, 'Test Man', 'COMP')

Error report -

ORA-02290: check constraint (COMP3311STU000.CHK_COURSEID) violated

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*** 16. CHECK CONSTRAINT on Course table: course id must be exactly four uppercase letters, a space and four digits.

Error starting at line : 102 in command -

insert into Course values ('COMP3311', '** Course Table: CHK: courseId', 3, 'Test Man', 'COMP')

Error report -

ORA-02290: check constraint (COMP3311STU000.CHK_COURSEID) violated

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More Details :

<https://docs.oracle.com/error-help/db/ora-02290/>

*** 17. CHECK CONSTRAINT on EnrollsIn table: grade must be between 0 and 100.

Error starting at line : 107 in command -

insert into EnrollsIn values ('99987654', 'COMP3311', 100.1)

Error report -

ORA-02290: check constraint (COMP3311STU000.CHK_GRADE) violated

<https://docs.oracle.com/error-help/db/ora-02290/>

More Details :

<https://docs.oracle.com/error-help/db/ora-02290/>

<<<<----- TEST FOR UNIQUE CONSTRAINT ----->>>>

*** 18. CHECK CONSTRAINT on Student table: email must be unique.

Error starting at line : 114 in command -

insert into Student values ('10101010', '*** Student table:', 'CHK: eEmail', 'csgrande', '22223334', '2023', 0.00, 'COMP')

Error report -

ORA-00001: unique constraint (COMP3311STU000.UNIQUE_EMAIL) violated

<https://docs.oracle.com/error-help/db/ora-00001/>

More Details :

<https://docs.oracle.com/error-help/db/ora-00001/>